

Vaibhav Anil Kahalekar

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Experience

IIT Hyderabad | Research Intern (MEGHA Group)

Feb 2026 – Jun 2026

- Conducted research on the Remote Sensing of Hydro-Meteorology within the **MEGHA research group** (Civil Engineering & Climate Change) under the guidance of Dr. Shruti Upadhyaya.
- Developed deep learning models to disentangle complex interactions between aerosols and precipitation in urban regions, utilizing high-dimensional satellite remote sensing observations.
- Processed large-scale hydro-meteorological datasets to identify atmospheric patterns, contributing to improved accuracy in urban climate modeling and forecasting.

Edudingo | Manual Testing Intern

Sep 2025 - Dec 2025

- Executed comprehensive **manual testing lifecycles** for core website features, documenting critical bugs and usability issues to streamline the deployment process for the development team.
- Collaborated with the engineering team to refine website content and UI logic, while proactively mastering web fundamentals to improve defect reporting accuracy.

Projects

CareerConnect | Python, Flask, Django Rest Framework, PostgreSQL, DeepSeek AI

- Architected a scalable backend recruitment platform using **Flask** and **Django** that leverages **OpenRouter (DeepSeek)** to parse unstructured resumes and extract technical skill sets with high precision.
- Designed a semantic search API using **Sentence-Transformers** to generate vector embeddings, ranking candidates against job descriptions via cosine similarity for context-aware matching.
- Built a "Referral Intelligence" system utilizing web scraping APIs to implement a priority algorithm that ranks connections to maximize referral success.

Time Series Sales Forecasting | Python, SARIMA, statsmodels, Data Analysis

- Conducted rigorous analysis on Perrin Freres Champagne sales data using **Augmented Dickey-Fuller (ADF)** tests and first-order differencing to isolate trends and achieve stationarity.
- Designed and fine-tuned a **SARIMA(1,1,1)x(1,1,1,12)** model based on **ACF/PACF** plots, successfully capturing complex 12-month seasonality where standard ARIMA models failed.
- Generated 12-month future sales forecasts with confidence intervals and validated model performance using residual diagnostics.

Technical Skills

Backend & Frameworks: Django, Django REST Framework, Flask, API Design, System Architecture

Databases & Cloud: PostgreSQL, AWS (EC2, S3)

Programming Languages: Python, C, C++, HTML, CSS

AI & Machine Learning: Large Language Models (LLMs), Prompt Engineering, Scikit-learn, Pandas, NumPy, PyTorch

Tools: Git, GitHub, Jupyter Notebook, VS Code, Linux

Relevant Coursework

Academic: Data Structures and Applications, Foundations of Machine Learning, Operating Systems, Computer Aided Numerical Methods, Database Management Systems, Computer Architecture and Organization

Certifications: Supervised Machine Learning: Regression and Classification (Coursera), Advanced Learning Algorithms(Coursera), C++ Programming Masterclass (Udemy)

Positions of Responsibility

TechLife Club Coordinator | Environmental Club, MGM Nanded

- Spearheaded environmental conservation initiatives by organizing tree plantation drives and integrating technology to monitor and promote wildlife protection awareness.

Education

MGM COE Nanded | Bachelor of Technology in Information Technology

2022 – 2026